

Literature answer for arXiv:1807.01355
Modular semilattice orthoscheme complexes are CAT(0)

Status. Literature already answered. Bacak’s survey *Old and new challenges in Hadamard spaces* asks, in the section on submodular functions on modular lattices, whether the orthoscheme complex $K(\mathcal{L})$ of every finite-rank modular semilattice $\mathcal{L}$ is a CAT(0) space. The survey states that this is Chalopin–Chepoi–Hirai–Osajda Conjecture 7.3.

Source problem. After recalling that Chalopin–Chepoi–Hirai–Osajda proved the result for finite-rank modular lattices, Bacak records the problem:

Let $\mathcal{L}$ be a modular semilattice of finite rank. Is $K(\mathcal{L})$ a CAT(0) space?

Answering paper. Hirai’s paper *A Nonpositive Curvature Property of Modular Semilattices* explicitly identifies the same conjecture. In the introduction it says that Chalopin et al. conjectured the CAT(0) property for modular semilattices and adds “see also [Problem 6.10]” in Bacak’s survey. Its main theorem is:

If $\mathcal{L}$ is a modular semilattice of finite rank, then $K(\mathcal{L})$ is CAT(0).

This is Theorem 1.2 in the arXiv version, and it is the same statement as the source problem.

Scope. This packet clears only this orthoscheme-complex question from the source survey. It does not address the survey’s other Hadamard-space problems, such as weak topologies, compactness of convex hulls, stochastic proximal point algorithms without local compactness, or asymptotic regularity of cyclic projections.

References

- [1] Miroslav Bacak. *Old and new challenges in Hadamard spaces*. arXiv:1807.01355, 2018.
- [2] Jeremie Chalopin, Victor Chepoi, Hiroshi Hirai, and Damian Osajda. *Weakly modular graphs and nonpositive curvature*. arXiv:1409.3892; *Memoirs of the AMS*.
- [3] Hiroshi Hirai. *A Nonpositive Curvature Property of Modular Semilattices*. arXiv:1905.01449; *Geometriae Dedicata* 214 (2021), 427–463.